

Vishal Munday

Applied AI Engineer | Backend Tech Lead | Voice AI

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Summary

Applied AI Engineer and Backend Tech Lead with 3+ years building production LLM agents, real-time voice systems, and backend infrastructure. Hands-on with end-to-end STT → LLM → tools → TTS pipelines, multi-turn state, WebRTC/LiveKit, RAG, provider integrations, and production AI workflows. Focused on latency, reliability, failure recovery, and measurable agent improvements.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL

AI/ML: LLM Agents, LangChain, LangGraph, RAG, Prompt Engineering, Tool Calling, Structured Outputs, Guardrails, Agent State, Mem0

Voice AI: STT/ASR, TTS, Voice Agents, Real-Time Audio, WebRTC, LiveKit, Streaming, Interruption Handling

Backend: Node.js, Express.js, REST APIs, WebSockets, Microservices, Async Processing

Data/Infra: PostgreSQL, MySQL, MongoDB, Redis, Prisma, Docker, AWS, GitHub Actions, Linux, CI/CD

Experience

EdgeNRoots Solutions Pvt Ltd

May 2024 – Present

Backend Tech Lead / Applied AI Engineer

Remote

- Lead backend and applied AI initiatives, owning architecture and production delivery for AI, SaaS, messaging, insurance, and automation workflows.
- Architected production voice-to-voice AI agents connecting STT, LLM reasoning, tool/function calling, conversation state, TTS, and real-time communication infrastructure.
- Built an end-to-end Meta voice/chat sequence integrating messaging events, agent orchestration, tools, state management, and response delivery for customer interactions.
- Designed agent behavior using prompts, structured outputs, tool calling, guardrails, error recovery, and deterministic business rules around LLM decisions.
- Evaluated STT/TTS and LLM providers across latency, streaming, concurrency, quality, reliability, and operational constraints for production voice systems.
- Investigated production agent failures across model output, tool execution, state, integrations, and latency; converted recurring failure patterns into prompt and engineering fixes.
- Built AI-powered approval, reconciliation, and automation workflows combining LLM reasoning with backend services, Redis, queues, and PostgreSQL/MySQL.
- Architected MetaOS, a WhatsApp CRM with campaigns, approvals, analytics, AI automation, and third-party integrations; reduced manual customer-response effort by 60%.
- Led backend development and mentored engineers across APIs, microservices, Dockerized deployments, CI/CD, and cloud infrastructure.

Icomply Lifescience Solution Pvt Ltd

Oct 2022 – Jun 2024

Full Stack Developer (Intern + Part-Time)

Remote

- Built backend APIs and web applications for B2B workflows; integrated AS2/XML-based secure data exchange for life-science clients.
- Optimized backend queries and application workflows, improving reliability and reducing average response times by 30%.

Selected Projects

Production Voice AI Agent • Python, Node.js, LiveKit, WebRTC, STT, LLM, TTS

- Built a real-time voice pipeline with streaming STT/LLM/TTS, tool execution, conversation state, interruption recovery, provider fallbacks, and production deployment.

FabricatorOS • Node.js, Express.js, PostgreSQL, Prisma, Docker, React – Manufacturing SaaS for quotations, product configuration, work orders, pricing, live SVG rendering, and production workflows.

ALMA Network – Alumni Platform • MERN, Firebase, AWS S3 – Alumni platform with profiles, authentication, media storage, community features, and scalable backend APIs.

Education

BITS Pilani (WILP)

B.Tech. in Artificial Intelligence and Machine Learning

2026 – Ongoing

Pilani, India

Guru BrahmaNand Ji Government Polytechnic

Diploma in Computer Engineering

2021 – 2024

Karnal, Haryana

Jawahar Navodaya Vidyalaya (JNV)

School Education